4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25

Question Count: 5

Code of Conduct

I K.Maheshbabu (192525122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Maheshbabu

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

1.

Cloud Architecture for Video streaming platform:

- use cloud storage to store video files and media content.
- Deploy content delivery network (CDN) to cache videos at edge locations closer to users.
- use load balancers to distribute user requests across multiple streaming servers.

Scalability and caching Techniques:

- use horizontal scaling by adding more streaming servers during peak traffic.
- CDN servers caches frequently wanted videos, reducing latency and buffering.
- Store popular content in memory caches for faster memory.

Role of load balancing and Auto-scaling:

load balancer:

- Distributes incoming traffic evenly among servers.
- Prevents any single server from becoming over-loaded.

Auto-scaling:

- Automatically adds servers when traffic evenly among server demand increases
- Removes unused servers during low traffic Periods

4. Cost optimization usage:

- use Pay-as-you-go cloud Services to avoid unnecessary infrastructure Costs.
- Cache frequently accessed Videos through CDN to reduce bandwidth Expenses.
- use auto-Scaling to run only the required number of Servers.

2. Multi-cloud Container Deployment for a fintech Startup:

Proposed Solution using Container orchestration and Multi-cloud:

- user Docker containers to package applications with all dependencies.
- Deploy Containers using Kubernetes as the Container orchestration platform.

Efficient Service Discovery:

- Kubernetes provides builtin-Service Dictionary through DNS Service.
- Each microserve recieves a unique DNS name, allowing Containers to Communicate without knowing Ip addresses.
- use Kubernettes Services and Ingress Controllers to route traffic efficiently.

Efficient scaling:-

- use Horizontal Pod Auto Scaler (HPA) to automatically increase (or) decrease container instances based on CPU, memory (or) custom metrics.

- use Cluster Autoscaler to add (or) remove worker needs nodes when workload changes

Benefits of Multi-Cloud deployment:-

- High avaliability failure of one cloud Provider does not affect the Entire application.

- Reduced Vendor risk: Application are not dependent on a single Provider.

Risks of Multi-cloud Deployment:-

- Managing multiple cloud Environments require additional Expertise.

- Monitoring and Management tools may increase Expenses.

- Consistent Security Policies must be maintained across Providers.

- Communication between clouds can introduce delays.

- Sychronizing data across clouds can be difficult

Bussiness continuity during failures:

- Create a detailed Disaster recovery (DR) plan.
- Deploy applications across multiple avaliability zones (or) regions.
- Implement automatic failover to secondary of infrastructure.

Benefits of the Proposed Solution:

- Prevents Permanent data loss from accidental deletion.
- Ensures quick recovery of Educational content and Student work.

4. Hybrid cloud Architecture for Government agency:

Hybrid Cloud Architecture design:

- Store Sensitive data and Critical application in the non-premises data Center to meet the Compliance requirements.
- Host non-sensitive applications and Public server in the Public cloud.

Secure Communication between Environments:

- Implement Site-to-site VPN or dedicated Private network links.
- use Secure Communication Protocols such as HTTPs and TLs/ssl.

3. Cloud backup and disaster Recovery Solution:-

1. solution using Cloud-Native backup, Replication, & Versioning:-

· Enable automatic Cloud backups for databases, application data, and learning Content.

· use across-region application to maintain copies of data in multiple geographic locations.

. Enable object Versioning for files and documents so deleted coo modified files can be restored easily.

2. Achieving RTo and RPo:-

· Recovery Time objective:-

· Maximum acceptable time to restore services after a failure.

· use automated failover mechanisms.

· Maintain Standby Servers in another region

· Recovery Point objective:-

· Maximum acceptable amount of data loss.

· use continuous data replication between the regions.

· Schedule frequent backups and database of Sychronization.

Identify Management Implementation:

- use a Centralized identity and Access Management (IAM) System.
- Enable Single-Sign-On (SSO) for seamless access across Environments.

Encryption Strategy:

- Encrypt data at rest using strong Encryption Standards such as AES-256.
- Encrypt data in transit using TLS/SSL protocols.
- use secure key management Services to store & manage Encryption keys.

Challenges in Managing Hybrid Cloud Environments:

- Maintaining resources across multiple Environments is difficult.
- Consistent Security Policies must be enforced everywhere.
- Regulatory rules may Very for different data-types.
- Connecting cloud and On-Premises application Can be Complex.

Multi-Region deployment strategy for a SaaS Provider:

Multi-Region deployment Strategy:

- Deploy the application in multiple cloud regions.
- Run application Servers, databases, and storage devices in both regions.
- use a Global load balancer to distribute user traffic to the nearest healthy region.

Data Replication Configuration:

- Enable real-time database replication b/w regions.
- use Sychronous replication for critical data requiring zero data loss.
- use Sychronous replication for less Critical workloads to reduce latency.

failover Mechanisms:

- Configure automatic failover through DNS-based or load-balancer-based routing
- Continuously Perform health Checks on application Servers and databases
- If the primary region becomes unavailable, traffic is automatically redirected to the secondary region

Monitoring and Allrting Strategies:

- Monitor cpu memory network, usage, database Performance, and application response time.
- use Centralized monitoring dashboards for all regions.
- Configure real-time alerts for:
  - Serve failures
  - High resources utilization
  - Database replication delays